

Gate cost and achievable parallelism of a momentum-conserving, two-body real UCC layer for the Hubbard model

Analysis note

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Abstract

We count the gates needed to compile the two-body, real (orthogonal), momentum- and spin-conserving unitary coupled-cluster layer of the Hubbard manuscript into a circuit, using the Trotterised product form and the Jordan–Wigner (JW) encoding with the standard CNOT-staircase construction of Barkoutsos *et al.* [1], and we ask how much of that cost can be removed by running independent generators concurrently.

The three main results are analytic and are confirmed by exhaustive enumeration. **(i)** The exact number of independent generators is $P = \frac{1}{4}(3N^3 - 6N^2 + (g+2)N)$ with $g = \prod_d \gcd(2, L_d)$, which is *smaller* than the $\frac{1}{4}(3N^3 + 2N^2)$ quoted in the manuscript; the difference is exactly the set of diagonal generators, which vanish identically, together with an implicit assumption that exactly one lattice dimension is even. The corrected formula reproduces the 114 parameters actually used in the 3×2 runs. **(ii)** Momentum conservation forces all four spin-orbitals of every generator to be distinct, so every generator produces exactly eight Pauli strings, of mean weight $\bar{w} = (5N + 24)/9$ and maximum weight $2N$. The total cost is $\frac{20}{3}N^4 + \frac{28}{3}N^3 - 40N^2 - 4(g+2)N$ CNOTs, $8P$ parametrised R_z rotations and $64P$ single-qubit Cliffords; for the 4×3 cluster of the manuscript this is 1.50×10^5 CNOTs on 24 qubits. **(iii)** Parallelism does not help. Because each generator’s JW support covers a fraction $\bar{w}/2N \rightarrow 5/18$ of the register, at most $18/5 = 3.6$ generators can ever run concurrently, and a greedy schedule achieves only a $1.23\times$ reduction in depth. The circuit depth remains $\Theta(N^4)$; there is no asymptotic parallel speedup, confirming the premise of the question.

We benchmark this against the depth counting of Cade *et al.* [13] for the efficient Hamiltonian Variational ansatz on the same model, after reconciling the two cost models. Their whole circuit reaches fidelity 0.99 in depth $\Theta(N^{3/2})$ — 108 two-qubit layers at 4×3 — against 1.24×10^5 for a *single* UCC layer, a factor of 1100. The reason is exactly the occupancy fraction above: their generators have support $w = 2$ and occupancy $1/N$, so $\Theta(N)$ of them run concurrently, where ours have occupancy $5/18$ and at most 3.6 do.

We then survey four routes to a cheaper circuit and estimate each. Balanced parity trees alone reduce depth from $\Theta(N^4)$ to $\Theta(N^3 \log N)$ at no gate cost; the excitation circuits of Yordanov *et al.* [2] reduce the CNOT count by a factor of 8 and the depth by $\approx 7.5\times$; and a doubly-factorised (unitary cluster Jastrow) *compilation* of the same amplitudes [10, 12, 9] reduces the depth by a factor of N . We separate two claims that are often run together: at full rank the decomposition is an identity (Proposition 4) and the compilation costs $\Theta(N^4)$ gates in $\Theta(N^3)$ depth unconditionally — $14\times$ shallower than the staircase at 4×3 at a *lower* gate count; compressed to rank $k = N$ it would cost $\Theta(N^3)$ gates in $\Theta(N^2)$ depth, but whether the manuscript’s $A^{(2)}$ admits that compression is an empirical question and is the experiment we recommend. Throughout, this route is a *compilation* of already-optimised coefficients, not a replacement ansatz: a single UCJ factor is indeed restricted to rotated density–density operators, but the sum over factors with non-commuting rotations is not, and is exact at rank $O(N^2)$.

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1 Scope and conventions

1.1 What is being compiled

The layer of the manuscript is

$$\mathcal{U}_A = \prod_{l=1}^L \prod_{(IJ)} \exp(A_{IJ}^{(l)} \hat{\tau}_{IJ}), \quad \hat{\tau}_{IJ} = \hat{C}_I^\dagger \hat{C}_J - \text{h.c.}, \quad (1)$$

with $I = (\mathbf{k}_3\sigma_3, \mathbf{k}_4\sigma_4)$, $J = (\mathbf{k}_1\sigma_1, \mathbf{k}_2\sigma_2)$ two-body multi-indices, i.e.

$$\hat{\tau}_{IJ} = \hat{c}_{\mathbf{k}_3\sigma_3}^\dagger \hat{c}_{\mathbf{k}_4\sigma_4}^\dagger \hat{c}_{\mathbf{k}_2\sigma_2} \hat{c}_{\mathbf{k}_1\sigma_1} - \text{h.c.}. \quad (2)$$

Three points fix the scope of what follows.

- **Product form, not the exact exponential.** Equation (1) is already a first-order Trotter product with one angle per generator. This is the only one of the two forms used in the manuscript that *is* a circuit: the single matrix exponential $e^{\sum_g A_g \hat{\tau}_g}$ has no direct gate realisation and would itself have to be Trotterised before compilation. All costs below are therefore for one Trotter layer, $L = 1$; multiply by L for deeper layers.
- **Real amplitudes.** The targets are real eigenstates of a real Hamiltonian, so $A^{(2)}$ is taken real symmetric rather than complex Hermitian. This is worth a factor of two: a complex Hermitian $A^{(2)}$ has $d(d-1)$ real off-diagonal degrees of freedom per momentum block against $d(d-1)/2$ for a real symmetric one, so the reality restriction already halves both P and the entire gate count quoted in this note (Remark 3).

- **Two-body only, $p = 2$.** As noted in the manuscript, the only momentum- and spin-conserving one-body generator is $\hat{n}_{\mathbf{k}\sigma}$, which is Hermitian and therefore gives $\hat{\tau} = 0$. The layer genuinely starts at $p = 2$.

1.2 Encoding and circuit model

We use $M = 2N$ qubits under the Jordan–Wigner transform, with the *spin-blocked* ordering

$$q(\mathbf{k}, \sigma) = \sigma N + \iota(\mathbf{k}), \quad \sigma \in \{\uparrow, \downarrow\} \equiv \{0, 1\}, \quad (3)$$

where $\iota : \text{BZ} \rightarrow \{0, \dots, N-1\}$ is any fixed enumeration of the momenta. Section 5.4 shows that the choice of ι changes the total cost by less than 3%, and that the alternative spin-interleaved ordering $q = 2\iota(\mathbf{k}) + \sigma$ is about 12% worse, so nothing is lost by fixing (3).

The cost model is the standard one: a Pauli exponential $e^{i\theta P}$ with $|\text{supp}(P)| = w$ is compiled as a basis change on the non- Z qubits, a CNOT ladder of length $w-1$, one R_z , and the inverse ladder and basis change — $2(w-1)$ CNOTs, one R_z and $2n_{XY}$ single-qubit Cliffords. We count CNOTs and CNOT depth, treat single-qubit gates as free for depth purposes, and assume all-to-all connectivity (Section 9.2 discusses what linear connectivity costs).

2 The generator set

2.1 Constraints

Write $\hat{\tau}_{IJ}$ with I the creation multi-index and J the annihilation multi-index. The constraints imposed in the manuscript are:

- (C1) **Fermionic antisymmetry.** I and J are unordered pairs of distinct spin-orbitals.
- (C2) **Spin.** $N_\uparrow$ and $N_\downarrow$ are separately conserved, so the spin multiset of I equals that of J . This leaves three channels: $\uparrow\uparrow \rightarrow \uparrow\uparrow$, $\downarrow\downarrow \rightarrow \downarrow\downarrow$ and $\uparrow\downarrow \rightarrow \uparrow\downarrow$.
- (C3) **Momentum.** $\mathbf{k}_3 + \mathbf{k}_4 \equiv \mathbf{k}_1 + \mathbf{k}_2 \equiv \mathbf{Q} \pmod{\mathbf{L}}$, i.e. I and J carry the same total momentum $\mathbf{Q}$.
- (C4) **Hermiticity/realty.** $A^{(2)}$ is real symmetric in (I, J) , and $\hat{\tau}_{IJ} = -\hat{\tau}_{JI}$, so only $I \neq J$ with an ordering convention (say $I \succ J$) are independent.

Consequently the generator set factorises over the N momentum sectors $\mathbf{Q}$, and within each sector over the three spin channels.

2.2 Everything is a genuine four-index excitation

Lemma 1 (Distinctness). *Under (C1)–(C3), every surviving generator $\hat{\tau}_{IJ}$ with $I \neq J$ involves four distinct spin-orbitals.*

Proof. Suppose I and J share a spin-orbital. In the same-spin channels, $I = \{\mathbf{k}_a, \mathbf{k}_b\}$, $J = \{\mathbf{k}_a, \mathbf{k}_c\}$ with all momenta at the same σ ; (C3) gives $\mathbf{k}_a + \mathbf{k}_b \equiv \mathbf{k}_a + \mathbf{k}_c$, hence $\mathbf{k}_b \equiv \mathbf{k}_c$ and $I = J$. In the opposite-spin channel, $I = (\mathbf{k}_a \uparrow, \mathbf{k}_b \downarrow)$ and $J = (\mathbf{k}_c \uparrow, \mathbf{k}_d \downarrow)$ can only share the $\uparrow$ orbital or the $\downarrow$ orbital; $\mathbf{k}_a = \mathbf{k}_c$ forces $\mathbf{k}_b \equiv \mathbf{k}_d$ by (C3), and symmetrically. Either way $I = J$, which is excluded by (C4). $\square$

Remark 1. Lemma 1 is what makes the gate count clean: without momentum conservation the generator set would contain number-operator-dressed single excitations $\hat{c}_p^\dagger \hat{n}_q \hat{c}_r$, which compile to four Pauli strings rather than eight and would have to be counted separately. Momentum conservation removes them entirely, so *every* generator costs exactly the same eight Pauli strings, and the only variation across the set is the JW string length.

Remark 2. Lemma 1 also means the diagonal $I = J$ terms all vanish: for same-spin, $\hat{c}_p^\dagger \hat{c}_q^\dagger \hat{c}_q \hat{c}_p = \hat{n}_p \hat{n}_q$ is Hermitian, so $\hat{\tau}_{II} = 0$, and identically in the opposite-spin channel. They must therefore be excluded from any parameter count. This is the source of the discrepancy in Section 3.5.

3 Parameter counting

3.1 Setup

Let $\delta_{\mathbf{Q}} = \#\{\mathbf{k} : 2\mathbf{k} \equiv \mathbf{Q}\}$ and note $\sum_{\mathbf{Q}} \delta_{\mathbf{Q}} = N$, since each $\mathbf{k}$ contributes to exactly one $\mathbf{Q} = 2\mathbf{k}$. On a rectangular cluster $L_1 \times L_2 \times \dots$ the condition factorises over dimensions, and $2k_d \equiv Q_d \pmod{L_d}$ has $\gcd(2, L_d)$ solutions when solvable and none otherwise, so

$$\sum_{\mathbf{Q}} \delta_{\mathbf{Q}}^2 = N g, \quad g \equiv \prod_d \gcd(2, L_d) = 2^{\#\{d: L_d \text{ even}\}}. \quad (4)$$

So $g = 1$ for an all-odd cluster (3×3 , 5×3), $g = 2$ for a mixed one (4×3 , 3×2), and $g = 4$ for an all-even one (4×4 , 6×2). This factor is the only shape dependence in the whole count.

3.2 Same-spin channels

Fix a spin σ and a total momentum $\mathbf{Q}$. The admissible index pairs are the unordered pairs $\{\mathbf{k}_1, \mathbf{k}_2\}$ with $\mathbf{k}_1 \neq \mathbf{k}_2$ and $\mathbf{k}_1 + \mathbf{k}_2 \equiv \mathbf{Q}$. There are exactly N ordered solutions of $\mathbf{k}_1 + \mathbf{k}_2 \equiv \mathbf{Q}$, of which $\delta_{\mathbf{Q}}$ are diagonal, so

$$d_{\mathbf{Q}} = \frac{N - \delta_{\mathbf{Q}}}{2}, \quad \sum_{\mathbf{Q}} d_{\mathbf{Q}} = \frac{N(N-1)}{2}, \quad \sum_{\mathbf{Q}} d_{\mathbf{Q}}^2 = \frac{N^3 - 2N^2 + gN}{4}, \quad (5)$$

the last line using (4). With a real symmetric $A^{(2)}$ restricted to $I \succ J$, the number of independent generators in one same-spin channel is $\sum_{\mathbf{Q}} \binom{d_{\mathbf{Q}}}{2}$:

$$n_{\sigma\sigma} = \frac{1}{2} \left(\sum_{\mathbf{Q}} d_{\mathbf{Q}}^2 - \sum_{\mathbf{Q}} d_{\mathbf{Q}} \right) = \frac{N^3 - 4N^2 + (g+2)N}{8}. \quad (6)$$

Both same-spin channels together contribute $2n_{\sigma\sigma}$.

3.3 Opposite-spin channel

Here $I = (\mathbf{k}_a \uparrow, \mathbf{k}_b \downarrow)$ is an *ordered* pair (one orbital of each spin), so for each $\mathbf{Q}$ there are exactly $d'_{\mathbf{Q}} = N$ configurations, labelled by $\mathbf{k}_a$ with $\mathbf{k}_b \equiv \mathbf{Q} - \mathbf{k}_a$ determined. Hence

$$n_{\uparrow\downarrow} = \sum_{\mathbf{Q}} \binom{N}{2} = \frac{N^2(N-1)}{2}. \quad (7)$$

3.4 Result

Proposition 1. *The number of independent real parameters of the two-body, momentum- and spin-conserving, real UCC layer is*

$$P = \frac{N^3 - 4N^2 + (g+2)N}{4} + \frac{N^2(N-1)}{2} = \frac{3N^3 - 6N^2 + (g+2)N}{4} \quad (8)$$

with $g = 2^{\#\{d: L_d \text{ even}\}}$. It is independent of the filling and, apart from g , of the cluster shape.

This was verified by exhaustive enumeration of the generator set for every cluster up to 8×6 ; Table 1 shows a representative subset. Asymptotically $P = \frac{3}{4}N^3 + O(N^2)$, in agreement with the $\frac{3}{4}N^3$ scaling quoted in the manuscript, with $n_{\uparrow\downarrow}/P \rightarrow 2/3$ and $2n_{\sigma\sigma}/P \rightarrow 1/3$.

Remark 3 (Reality). If $A^{(2)}$ were complex Hermitian instead of real symmetric, each off-diagonal block entry would carry two real parameters and each generator would split into $\hat{C}_I^\dagger \hat{C}_J - \text{h.c.}$ and $i(\hat{C}_I^\dagger \hat{C}_J + \text{h.c.})$. P , and with it every gate count in this note, would exactly double. The restriction to real orthogonal $\mathcal{U}_A$ is therefore already a factor-of-two saving in circuit cost, not only in optimisation cost.

3.5 Comparison with the formula in the manuscript

Appendix D of the manuscript quotes $P_{\text{ms}} = \frac{1}{4}(3N^3 + 2N^2)$ for even N . This is larger than (8), and the difference is exactly accounted for:

$$P_{\text{ms}} - P = \underbrace{\frac{2N^2 - N}{\text{diagonal } I=J \text{ terms}}} + \underbrace{\frac{1}{4}(2-g)N}_{\text{shape}}. \quad (9)$$

Two things have happened.

1. The manuscript’s recipe counts $d(d+1)/2$ independent entries of a real symmetric block, i.e. it *includes the diagonal*. But $\hat{\tau}_{II} = 0$ identically, so those $2N^2 - N$ “parameters” generate nothing. They inflate P without changing the ansatz.
2. The remaining $\frac{1}{4}(2-g)N$ says the quoted formula silently assumes $g = 2$, i.e. exactly one even lattice dimension. It is correct (given point 1) for 4×3 and 3×2 , but off by -6 for 6×2 and -8 for 4×4 .

The corrected formula (8) agrees with what the code actually does. At $N = 6$ (3×2 , $g = 2$) it gives $P = 114$, which is the number of parameters reported for the 3×2 ($3, 2$) head-to-head runs, against 180 from the quoted formula; the code evidently drops the vanishing diagonal generators. At $N = 12$ the two agree numerically only by coincidence of the 4×3 shape: $P = 1092$ versus 1368 quoted.

Table 1: Generator counts by channel, from exhaustive enumeration. P_{ms} is the formula quoted in the manuscript. The last column is the excess $P_{\text{ms}} - P$, which is $2N^2 - N$ (the identically vanishing diagonal generators) plus the shape term $\frac{1}{4}(2 - g)N$; the two coincide only for $g = 2$.

cluster	N	$M = 2N$	g	$2n_{\sigma\sigma}$	$n_{\uparrow\downarrow}$	P (Eq. 8)	P_{ms}	$P_{\text{ms}} - P$
2×2	4	8	4	6	24	30	56	26
3×2	6	12	2	24	90	114	180	66
4×2	8	16	4	76	224	300	416	116
3×3	9	18	1	108	324	432	585	153
4×3	12	24	2	300	792	1092	1368	276
4×4	16	32	4	792	1920	2712	3200	488
5×4	20	40	2	1620	3800	5420	6200	780
6×4	24	48	4	2916	6624	9540	10656	1116
6×6	36	72	4	10422	22680	33102	35640	2538

4 Compiling one generator

4.1 Pauli decomposition

With $\hat{c}_j = \frac{1}{2}(X_j + iY_j) \prod_{r < j} Z_r$ and the four distinct spin-orbitals $q_i < q_j < q_k < q_l$ guaranteed by Lemma 1, the JW image of a double excitation is the textbook eight-term sum [1, 2],

$$\hat{\tau} \mapsto \frac{i}{8} \left(X_i Y_j X_k X_l + Y_i X_j X_k X_l + Y_i Y_j Y_k X_l + Y_i Y_j X_k Y_l \right. \\ \left. - X_i X_j Y_k X_l - X_i X_j X_k Y_l - Y_i X_j Y_k Y_l - X_i Y_j Y_k Y_l \right) \prod_{r=i+1}^{j-1} Z_r \prod_{r'=k+1}^{l-1} Z_{r'}. \quad (10)$$

All eight strings commute with one another, and — crucially for the depth analysis — all eight have the *same* support. Writing the sorted qubit indices of the four spin-orbitals as $q_{(1)} < q_{(2)} < q_{(3)} < q_{(4)}$, the Z -strings run over the *inner* gaps only (the number of indices exceeding q is odd precisely on $(q_{(1)}, q_{(2)})$ and $(q_{(3)}, q_{(4)})$), so the support has size

$$w = 4 + (q_{(2)} - q_{(1)} - 1) + (q_{(4)} - q_{(3)} - 1) = 2 + (q_{(2)} - q_{(1)}) + (q_{(4)} - q_{(3)}). \quad (11)$$

This w is exactly Yordanov's $n^{(df)} = j - i + l - k + 2$.

4.2 Cost of one generator

Compiling each of the eight strings as a CNOT ladder gives, per generator,

$$n_{\text{CNOT}} = 16(w - 1), \quad n_{R_z} = 8, \quad n_{1q}^{\text{Clifford}} = 64, \quad \text{depth} = 16(w - 1), \quad (12)$$

where the depth equals the count because the ladders are strictly sequential and the eight strings overlap completely. Equation (12) reproduces the $16(n^{(df)} - 1)$ quoted by Yordanov *et al.* for the standard construction [2], which is a useful independent check on the model.

Two observations that matter later. First, the R_z count $8P$ is the number of *parametrised* rotations, eight per variational angle; this is the quantity that sets the cost of parameter-shift gradients on hardware. Second, a balanced parity tree in place of the ladder [5] leaves the CNOT count unchanged but reduces the depth of one string from $2(w - 1)$ to $2\lceil \log_2 w \rceil$.

5 Weight statistics

The total cost is $16 \sum_g (w_g - 1)$, so everything reduces to the distribution of w over the generator set. This can be done analytically for the opposite-spin channel and to leading order for the same-spin channels.

5.1 Opposite-spin channel: exact

Proposition 2. *Under the spin-blocked ordering (3), the mean Pauli weight in the opposite-spin channel is*

$$\bar{w}_{\uparrow\downarrow} = \frac{2N + 8}{3}, \quad (13)$$

exactly, for every cluster shape and every momentum enumeration ι .

Proof. With the ordering (3), the two $\uparrow$ orbitals occupy qubits $a, c \in [0, N)$ and the two $\downarrow$ orbitals occupy qubits $N + b, N + d$. Since the two blocks are disjoint and ordered, $q_{(1)}, q_{(2)}$ are the $\uparrow$ pair and $q_{(3)}, q_{(4)}$ the $\downarrow$ pair, so (11) gives $w = 2 + |a - c| + |b - d|$.

Fix $\mathbf{Q}$. The configurations in that sector are labelled bijectively by the $\uparrow$ momentum, with the $\downarrow$ momentum determined. A generator is an unordered pair of two distinct configurations, so the induced $\uparrow$ index pair $\{a, c\}$ ranges over *all* $\binom{N}{2}$ unordered distinct pairs of $[0, N)$, once each; summing over the N sectors, each unordered pair occurs exactly N times. Hence $\{a, c\}$ is uniform on distinct pairs and

$$\mathbb{E} |a - c| = \frac{1}{\binom{N}{2}} \sum_{a < c} (c - a) = \frac{2}{N(N-1)} \sum_{h=1}^{N-1} h(N-h) = \frac{2}{N(N-1)} \cdot \frac{N(N^2-1)}{6} = \frac{N+1}{3}. \quad (14)$$

The whole generator set is invariant under interchanging the roles of $\uparrow$ and $\downarrow$, so $\{b, d\}$ has the same marginal distribution and $\mathbb{E} |b - d| = (N+1)/3$ as well. Therefore $\bar{w}_{\uparrow\downarrow} = 2 + 2(N+1)/3 = (2N+8)/3$. $\square$

Enumeration confirms this to the last digit for every cluster tested, e.g. $32/3$ at $N = 12$ and $80/3$ at $N = 36$.

5.2 Same-spin channels

Here all four qubits lie in one spin block and the four momenta are correlated by $\mathbf{k}_1 + \mathbf{k}_2 \equiv \mathbf{k}_3 + \mathbf{k}_4$, so the same-spin pairs are *not* uniform on 4-subsets. Numerically,

$$\bar{w}_{\sigma\sigma} = \frac{N+8}{3} + O(N^{-1}), \quad (15)$$

with a small, shape-dependent negative correction: $\bar{w}_{\sigma\sigma} = 20/3$ exactly at $N = 12$, $262/33 = 7.939$ against 8 at $N = 16$, and 18.639 against 18.667 at $N = 48$. The correction never exceeds 2% and is irrelevant at the level of a resource estimate.

5.3 Combined

Using $2n_{\sigma\sigma}/P \rightarrow 1/3$ and $n_{\uparrow\downarrow}/P \rightarrow 2/3$,

$$\boxed{\bar{w} = \frac{1}{3} \cdot \frac{N+8}{3} + \frac{2}{3} \cdot \frac{2N+8}{3} = \frac{5N+24}{9} + O(1).} \quad (16)$$

The maximum weight is $w_{\max} = 2N = M$: there are always generators whose JW string spans the entire register. Equation (16) is the single most consequential number in this note, because it says the *average* generator touches a fixed fraction

$$\frac{\bar{w}}{2N} \longrightarrow \frac{5}{18} \approx 0.278 \quad (17)$$

of the register. Enumerated values: 0.399 at $N = 12$, 0.367 at $N = 16$, 0.318 at $N = 36$, 0.308 at $N = 48$, converging slowly to $5/18$ from above.

5.4 Does the qubit ordering help?

It does not. For 4×3 , the spin-blocked ordering with the natural momentum enumeration gives $\bar{w} = 9.568$; the best of 4000 random momentum permutations gives 9.484 (-0.9%) and the worst 9.769 ($+2.1\%$). The spin-interleaved ordering $q = 2\iota(\mathbf{k}) + \sigma$ gives 10.762, i.e. 12% *worse*, because it doubles the qubit distance between the two members of each same-spin pair. The reason no ordering helps is structural: by Proposition 2 the $\uparrow$ index pair of the opposite-spin generators is uniform over *all* distinct pairs, so the mean gap $(N+1)/3$ is a permutation invariant. Optimising the JW ordering is not a lever here.

6 Total gate count

Combining P from (8) with the channel-resolved weights,

$$\sum_g w_g = 2n_{\sigma\sigma}\bar{w}_{\sigma\sigma} + n_{\uparrow\downarrow}\bar{w}_{\uparrow\downarrow} = \frac{5N^4 + 16N^3 - 48N^2}{12} + O(N^2), \quad (18)$$

and therefore

$$n_{\text{CNOT}} = 16 \left(\sum_g w_g - P \right) = \frac{20}{3}N^4 + \frac{28}{3}N^3 - 40N^2 - 4(g+2)N + O(N^2), \quad (19)$$

together with

$$n_{R_z} = 8P = 6N^3 + O(N^2), \quad n_{1q} = 64P = 48N^3 + O(N^2). \quad (20)$$

Equation (19) agrees with the exhaustive enumeration to 0.9% at $N = 12$ and 0.05% at $N = 36$ (Table 2). The headline scaling is

$$n_{\text{CNOT}} \simeq \frac{20}{3}N^4 \approx 6.7 N^4, \quad (21)$$

i.e. $\Theta(N^4)$ two-qubit gates for a layer with only $\Theta(N^3)$ parameters. The extra factor of N is entirely the Jordan–Wigner string.

The two clusters of the manuscript. For 4×3 — the largest system with fidelity data, sector dimension 32,670 — one layer costs 1.50×10^5 CNOTs, 8736 parametrised R_z and 6.99×10^4 single-qubit Cliffords on 24 qubits. For 4×4 it is 4.67×10^5 CNOTs on 32 qubits. For context, keeping the accumulated two-qubit error below unity at 4×3 requires a CNOT infidelity below 7×10^{-6} , roughly two to three orders of magnitude beyond current superconducting hardware. This is a fault-tolerant-era circuit as compiled, which is the quantitative form of the concern raised about Ref. [1].

Table 2: Exact gate counts and depths for one Trotter layer ($L = 1$), from exhaustive enumeration under the standard CNOT-staircase compilation. D_{seq} is the fully sequential CNOT depth, D_{par} the depth of a greedy disjoint-support schedule. The last column is the asymptotic formula (19).

cluster	N	M	P	$\bar{w}$	n_{CNOT}	n_{R_z}	D_{seq}	D_{par}	Eq. (19)
2×2	4	8	30	5.07	1.95×10^3	240	1.95×10^3	1.71×10^3	-19.7%
3×2	6	12	114	6.21	9.50×10^3	912	9.50×10^3	8.40×10^3	-4.0%
4×2	8	16	300	7.31	3.03×10^4	2400	3.03×10^4	2.57×10^4	-3.1%
3×3	9	18	432	7.89	4.76×10^4	3456	4.76×10^4	4.08×10^4	-0.9%
4×3	12	24	1092	9.57	1.50×10^5	8736	1.50×10^5	1.24×10^5	-0.9%
4×4	16	32	2712	11.76	4.67×10^5	21696	4.67×10^5	3.85×10^5	-0.5%
5×4	20	40	5420	13.99	1.13×10^6	43360	1.13×10^6	9.31×10^5	-0.1%
6×4	24	48	9540	16.21	2.32×10^6	76320	2.32×10^6	1.90×10^6	-0.2%
6×6	36	72	33102	22.88	1.16×10^7	264816	1.16×10^7	9.42×10^6	-0.1%

7 How much parallelism is actually available?

7.1 The scheduling problem

Two generators can be executed concurrently if and only if their qubit supports are disjoint. (Within a generator there is nothing to parallelise: by (10) all eight strings share a single support.) Scheduling the layer is therefore list-scheduling on the conflict graph whose vertices are the P generators and whose edges join overlapping supports. The generators commute only within a momentum sector, but since we are free to choose the Trotter ordering, any schedule that respects a fixed linear order is admissible.

7.2 An occupancy bound

The decisive constraint is (17). At any instant the generators running concurrently have pairwise disjoint supports, so their weights sum to at most $M = 2N$. Hence the instantaneous concurrency is at most $2N/\bar{w}$, and averaged over the layer the achievable speedup is bounded by

$$\boxed{\frac{D_{\text{seq}}}{D_{\text{par}}} \leq \frac{2N}{\bar{w}} \longrightarrow \frac{18}{5} = 3.6.} \quad (22)$$

This is a hard, N -independent ceiling. It is the central answer to the question: the Jordan–Wigner string makes every two-body generator a *global* operation on a constant fraction of the register, so no more than a handful can ever run at once, no matter how many qubits there are. Enumerated values of $2N/\bar{w}$: 2.51 at $N = 12$, 2.72 at $N = 16$, 3.15 at $N = 36$, 3.25 at $N = 48$.

7.3 What a greedy schedule achieves

The bound (22) is not tight, because the supports are far from randomly placed: every string covers an interval, and intervals cluster around the middle of each spin block. A greedy earliest-start schedule over all P generators (largest support first) gives

$$\frac{D_{\text{seq}}}{D_{\text{par}}} = 1.14, 1.21, 1.21, 1.23, 1.24 \quad \text{at } N = 6, 12, 16, 24, 48, \quad (23)$$

saturating at about 1.25. A second, schedule-independent lower bound confirms this: the busiest qubit alone forces $D_{\text{par}} \geq \max_q \sum_{g \ni q} d_g$, which already equals $D_{\text{seq}}/1.63$ at $N = 48$. The load profile is strongly peaked — for 4×4 the central qubit of each spin block carries $4.4\times$ the CNOT load of the edge qubits:

71k 135k 190k 235k 267k 289k 304k 311k | 311k 304k 289k 267k 235k 190k 135k 71k ($\uparrow$ block)

and identically for the $\downarrow$ block. The middle of the register is a hard bottleneck because it lies inside the Z -string of almost every generator.

Proposition 3 (No asymptotic parallel speedup). *Under the standard CNOT-staircase compilation with any Jordan–Wigner ordering, the two-body momentum-conserving UCC layer has parallel CNOT depth $D_{\text{par}} = \Theta(N^4)$, the same order as its gate count, with the constant reduced from $20/3$ to at best $20/(3 \cdot 3.6) = 1.85$ and in practice to ≈ 5.4 .*

7.4 What *does* reduce the depth for free

Replacing each CNOT ladder by a balanced parity tree [5] — explicitly suggested for these circuits by Yordanov *et al.* — leaves the gate count at (19) but reduces the depth of one generator from $16(w-1)$ to $16\lceil \log_2 w \rceil$. The layer depth then becomes

$$D_{\text{seq}}^{\text{tree}} = \Theta(N^3 \log N), \quad (24)$$

and the greedy schedule now buys $1.4\text{--}1.46\times$ on top (the blocks are shorter, so they pack better). At 4×3 this takes the parallel depth from 1.24×10^5 to 5.2×10^4 . This is the single cheapest improvement available and requires no change to the ansatz at all. Note the asymmetry it creates: depth drops by a factor of $N/\log N$ while the gate count is untouched, so the circuit becomes gate-count- rather than depth-limited.

8 Comparison with the depth counting of Cade *et al.*

Reference [13] performs the analogous exercise for the Hamiltonian Variational family on the same model, and reaches depths three to four orders of magnitude smaller. It is the natural benchmark for Section 7, because its ansatz is precisely the case in which the parallelism obstruction identified there is absent. It is also the ansatz already reimplemented in this project for the head-to-head runs, so the comparison is directly relevant.

8.1 Reconciling the two cost models

The two counts are not in the same units, and the difference must be removed before any ratio is meaningful.

- **Gate set.** Cade *et al.* charge depth 1 for an *arbitrary* two-qubit gate and zero for one-qubit gates, on the grounds that their native operation $e^{i(\theta(XX+YY)+\gamma ZZ)}$ is directly available on superconducting hardware. This note charges depth 1 per CNOT. A generic two-qubit gate costs up to three CNOTs, so their depths should be multiplied by ≈ 3 to be read as CNOT depths — as should their FSWAPS.
- **Why the converse correction is negligible.** One might ask whether our CNOT depths should conversely be *divided* by three, by merging adjacent CNOTs into arbitrary two-qubit gates.

They should not: consecutive CNOTs in a Jordan–Wigner ladder act on *different* qubit pairs $(q, q+1), (q+1, q+2), \dots$, so they do not merge. Our CNOT depth is already, to within a small constant, a two-qubit-gate depth in their units.

- **Connectivity.** We assumed all-to-all; the comparable row of their Table I is “Fully Connected”. Their nearest-neighbour and Sycamore rows are $2\times$ and $3\times$ worse, and our counts would degrade correspondingly under restricted connectivity (Section 9.2).
- **Encoding.** Both use Jordan–Wigner — they in real space with a snake ordering, we in momentum space. They report that JW is the best encoding for grids with $\min\{n_x, n_y\} \leq 8$, which covers every cluster considered here, so the comparison is not an artefact of a poor encoding choice on either side.

8.2 Their count

For the efficient Hamiltonian Variational (EHV) ansatz they obtain a depth per layer of

$$D_{\text{layer}}^{\text{EHV}} = \begin{cases} 2\min\{n_x, n_y\} + 1 & \min\{n_x, n_y\} \text{ even} \\ 2\min\{n_x, n_y\} + 2 & \min\{n_x, n_y\} \text{ odd} \end{cases} \quad (25)$$

on a fully connected architecture ($4n_x$ and $6n_x + 1$ for nearest-neighbour and Sycamore connectivity respectively). Preparing the non-interacting initial state by Givens rotations costs a further $N - 1$. Numerically they find 0.99 fidelity to the ground state with at most $1.5N$ layers, and $\sim N$ layers in most cases, giving a total

$$D^{\text{EHV}} = (N - 1) + n_{\text{layers}} D_{\text{layer}}^{\text{EHV}} + 1. \quad (26)$$

Their worked example is a 5×5 grid on 50 qubits with 25 layers: $24 + 25 \times 12 + 1 = 325$. Equation (26) as implemented here reproduces that number exactly, which fixes the convention.

For a square grid $\min\{n_x, n_y\} = \sqrt{N}$ and $n_{\text{layers}} = \Theta(N)$, so

$$D^{\text{EHV}} = \Theta(N^{3/2}), \quad (27)$$

against $\Theta(N^4)$ for the two-body momentum-space UCC layer. (On a ladder of fixed width the EHV depth is only $\Theta(N)$, since $\min\{n_x, n_y\}$ is then constant.)

8.3 Side by side

Expressivity per unit depth

Because Table 3 now carries both axes, the natural figure of merit — degrees of freedom bought per layer of circuit depth — can be read off its scaling row:

$$\frac{\text{DOF}}{\text{depth}} \sim \begin{cases} N/N^{3/2} = N^{-1/2} & \text{EHV} \\ N^2/N^{3/2} = N^{+1/2} & \text{NP} \\ N^3/N^4 = N^{-1} & \text{UCC, staircase or Yordanov} \\ N^3/N^2 = N^{+1} & \text{UCC, doubly factorised} \end{cases} \quad (28)$$

Two things have to be said about this ordering, and they pull in opposite directions.

Table 3: **Degrees of freedom against circuit depth.** Depths are in layers of two-qubit gates on a fully connected architecture; DOF is the number of independent variational parameters. **Left block:** the complete Cade *et al.* [13] circuit — state preparation plus $n_{\text{layers}} = N$ ansatz layers, their empirical requirement for fidelity 0.99. The EHV and NP ansätze differ in DOF but share a depth (both are “efficient ansätze” in their Table I), which is why one depth column serves both; it is Eq. (26). On the DOF side, EHV ties each of the five commuting Hamiltonian term-sets of their Eq. (2) to one angle, giving at most 5 per layer (fewer on narrow grids, where some sets are absent), while NP gives every mode pair its own two angles, $10N - 4(n_x + n_y)$ per layer. Their conservative bound $n_{\text{layers}} = 1.5N$ raises depths and DOF alike by 1.5, leaving DOF-per-depth unchanged. **Right block:** a *single* Trotter layer of the momentum-space ansatz, Eq. (1) with $L = 1$, whose DOF is P of Proposition 1 regardless of how it is compiled, under three compilations: the standard CNOT-staircase construction (Section 6), Yordanov excitation circuits (Section 9), and the doubly-factorised form at $k = N$ (Section 9.3). All are greedy parallel schedules, i.e. the parallelism of Section 7 is already included. Parenthesised values are that depth divided by the Cade depth in the same row. **Units:** staircase and Yordanov columns are CNOT depths; the Cade and UCJ columns are depths in arbitrary two-qubit gates, worth up to three CNOTs each, so multiply those by ≈ 3 for a strict comparison (Section 8.1). This changes no conclusion. **Note the two near-matched DOF pairs** (Section 8.5): NP on 3×2 at 480 against the manuscript’s own HVA runs at 464–480, and NP on 4×3 at 1104 against $P = 1092$.

		Cade <i>et al.</i> : whole circuit to $F = 0.99$				UCC: <i>one</i> Trotter layer		
		DOF		depth	DOF = P	depth (rel. to Cade)		
cluster	qubits	EHV	NP			staircase	Yordanov	DF _{$k=N$}
2×2	8	12	96	24	30	1.7×10^3 (71)	3.3×10^2 (14)	60 (2.5)
3×2	12	24	240	36	114	8.4×10^3 (233)	1.3×10^3 (37)	114 (3.2)
4×2	16	32	448	48	300	2.6×10^4 (536)	3.8×10^3 (79)	184 (3.8)
3×3	18	45	594	81	432	4.1×10^4 (503)	5.8×10^3 (72)	225 (2.8)
4×3	24	60	1104	108	1092	1.2×10^5 (1149)	1.7×10^4 (156)	372 (3.4)
4×4	32	80	2048	160	2712	3.9×10^5 (2407)	5.1×10^4 (318)	624 (3.9)
5×4	40	100	3280	200	5420	9.3×10^5 (4655)	1.2×10^5 (607)	940 (4.7)
6×4	48	120	4800	240	9540	1.9×10^6 (7898)	2.4×10^5 (1020)	1320 (5.5)
6×6	72	180	11232	504	33102	9.4×10^6 (18692)	1.2×10^6 (2383)	2844 (5.6)
<i>scaling</i>		$\Theta(N)$	$\Theta(N^2)$	$\Theta(N^{3/2})$	$\Theta(N^3)$	$\Theta(N^4)$	$\Theta(N^4)$	$\Theta(N^2)$

Asymptotically, the doubly-factorised UCC layer wins, and by the widest margin — it is the only one of the four whose expressivity per unit depth *grows* linearly in N . Note also that the $8\times$ improvement from Yordanov circuits does not move the ansatz on this metric at all, because it reduces a constant and not an exponent: the ordering is a statement about compilation strategy, not about the variational manifolds, which is why Section 9.3 rather than Section 9 carries the recommendation.

But the constants put the crossover far outside the range of interest. Evaluating DOF/depth on actual clusters:

cluster	EHV	NP	UCC staircase	UCC Yordanov	UCC DF ($k=N$)
4×3	0.6	10.2	0.009	0.065	2.9
4×4	0.5	12.8	0.007	0.053	4.3
6×6	0.4	22.3	0.004	0.028	11.6
10×10	0.2	41.8	0.001	0.011	35.5
14×14	0.2	61.6	0.001	0.006	71.5
20×20	0.1	91.4	—	—	148.0

NP remains ahead of the doubly-factorised UCC layer until $N \approx 150$, i.e. a 12×12 cluster and beyond — far past anything either paper computes, and past the point where the comparison is decided by other considerations entirely. On the clusters of interest the honest statement is that **NP buys expressivity per unit depth some 2–4× more cheaply than the best available compilation of the momentum-space layer**, and that the asymptotic ordering, while real, is not yet operative. What *is* operative on those clusters is that the momentum-space layer has 10–30× more degrees of freedom in absolute terms, and reaches infidelities nine orders of magnitude lower.

Absolute numbers

Table 3 is stark. At 4×3 — the largest cluster with fidelity data in the manuscript — a *single* UCC Trotter layer is 1100× deeper than an entire EHV circuit, or 160× after recompiling with Yordanov circuits, and even the doubly factorised form of Section 9.3 remains 8× deeper. The ratios grow with N , as they must: $\Theta(N^4)$ against $\Theta(N^{3/2})$.

8.4 Why: the occupancy fraction again

The gap is not an accident of compiler quality, and Section 7 already contains its explanation. The relevant quantity is the fraction of the register a single generator occupies.

	support w	occupancy $w/2N$	max concurrency
EHV hopping / onsite term	2	$1/N$	$\Theta(N)$
Momentum-space UCC generator	$(5N + 24)/9$	$\rightarrow 5/18$	≤ 3.6

The EHV generators are geometrically local in real space, and the snake ordering plus the FSWAP network of Ref. [8] keeps them Jordan–Wigner adjacent, so every term is a weight-2 operation and $\Theta(N)$ of them execute concurrently. That is why their layer depth is $O(\sqrt{N})$ rather than $O(N)$ times their gate count. Momentum-space UCC generators are, by construction, delocalised over the whole lattice; by Proposition 2 their mean JW support is a *constant fraction* of the register, capping concurrency at 3.6 for all N . Cade *et al.*’s numbers are therefore the empirical demonstration of the bound (22) being saturated at one end while our ansatz sits at the other.

It is worth adding that Ref. [13] paid for this: their EHV ansatz is *not* a term-by-term Trotterisation of the HV ansatz. They note that implementing the HV terms directly on a $2 \times n_y$ grid costs $2 + 2 \times (1 + 4) = 12$ per layer against their 5, because the vertical hopping terms then require $e^{i\theta(XX+YY)ZZ}$ at depth 4. The factor of $N/\log N$ available from balanced parity trees in Section 7.4 is the analogous — and much smaller — win available on our side.

8.5 What the comparison does and does not establish

Three caveats, all of which cut against reading Table 3 as a verdict on the ansatz.

1. **Different accuracy targets.** Cade *et al.* count the depth needed to reach fidelity 0.99. The manuscript reports infidelity 7.1×10^{-14} on 3×2 with a single layer. These are not the same question, and Table 3 compares a converged EHV circuit against one UCC layer, i.e. it is generous to EHV on layers and generous to UCC on accuracy.
2. **At matched degrees of freedom the depth gap barely narrows.** Two rows of Table 3 allow an almost DOF-matched comparison, and they are the two that matter.
 On 3×2 , the NP ansatz has $10N - 4(n_x + n_y) = 40$ parameters per layer, so 12 layers give 480 — exactly the 464–480 parameters reported for this project’s own HVA reimplementation with untied couplings. That identification is worth recording on its own: the `--hva_tie=none` runs are Cade *et al.*’s NP ansatz at twelve layers. Those twelve layers cost $12 \times 5 + 1 = 61$ two-qubit layers from a basis state, or 66 with Givens state preparation. One UCC layer on the same cluster, with 114 parameters, costs 8.4×10^3 — a factor of 130 more depth for a quarter of the DOF.
 On 4×3 the match is closer still: NP at 12 layers has 1104 parameters against $P = 1092$, i.e. DOF matched to 1%, at depth 108 against 1.2×10^5 (staircase), 1.7×10^4 (Yordanov) or 864 (UCJ). So the momentum-space ansatz is $1100\times$, $160\times$ or $8\times$ deeper *at equal parameter count*.
 None of this contradicts the manuscript’s results; it reframes them. The manuscript’s advantage is not DOF — at 4×3 the two ansätze have essentially the same number — but what those DOF achieve: 7.1×10^{-14} infidelity on 3×2 against 6.2×10^{-3} to 3.6×10^{-5} for HVA/NP at four times the parameters. The honest summary is that the two ansätze trade roughly nine orders of magnitude in infidelity against roughly two orders of magnitude in depth. Which side of that trade is preferable cannot be settled from either paper.
3. **The missing experiment.** Neither paper has the curve that would settle it: *circuit depth at matched infidelity, for both ansätze, on the same cluster*. The ingredients now exist in this project — the HVA is implemented in the same codebase, and Table 3 supplies the depth model for both sides. Plotting infidelity against compiled two-qubit depth (rather than against parameter count, as in Fig. 4 of the manuscript) would turn the cost–performance framing of that figure into a statement about hardware, and is the single most informative addition available. If the UCJ compression of Section 9.3 closes the remaining factor of 8 at 4×3 , the momentum-space ansatz becomes competitive on depth as well; if it does not, the honest conclusion is that this ansatz is a classical-structure result rather than a near-term circuit proposal, which is in any case how the manuscript currently frames its hardware outlook.

9 Cheaper implementations

Four routes, in increasing order of how much they change the ansatz.

9.1 Route 1 — better excitation circuits (no change to the ansatz)

The eight strings of (10) commute and differ only in their X/Y pattern, so compiling them as eight independent ladders is wasteful. Two standard fixes:

Yordanov–Arvidsson–Shukur–Barnes circuits [2]. Instead of eight CNOT staircases, they implement the double excitation as a parity-controlled exchange: two CNOTs encode the parity of

each pair, a multiply-controlled R_y performs the exchange, and a *single* CNOT staircase supplies the JW sign. Their explicit circuit has

$$n_{\text{CNOT}} = 2w + 5, \quad \text{depth} = \max[13, 2w - 1], \quad (29)$$

against $16(w - 1)$ for both in the standard construction — the factor of 8 they quote. Applied to our generator set this gives

$$n_{\text{CNOT}}^{\text{Yord}} = 2 \sum_g w_g + 5P = \frac{5}{6}N^4 + O(N^3), \quad (30)$$

and enumerated: 2.64×10^4 CNOTs at 4×3 (a $5.7\times$ reduction) and 7.73×10^4 at 4×4 ($6.0\times$). The greedy parallel depth falls to 1.69×10^4 at 4×3 , a $7.4\times$ reduction; with balanced trees on the one remaining staircase, 1.47×10^4 ($8.4\times$). A recent decomposition reduces the four-qubit core from 13 CNOTs and depth 11 to 12 and depth 8 [4], worth a further few percent. Magoulas and Evangelista extend the construction to arbitrary excitation rank and give approximate variants with up to a further $4\times$ CNOT reduction at negligible energy cost [3].

Simultaneous diagonalisation of commuting Pauli clusters [6, 7]. Since all eight strings of one generator commute, a single Clifford can diagonalise the whole cluster, after which the eight rotations are phase gadgets on a shared parity structure and can be synthesised jointly by phase-polynomial methods. Clusters can also be formed *across* generators within a momentum sector. This is what a good compiler (TKET, Qiskit’s Pauli-evolution passes) will do automatically and is largely complementary to the Yordanov circuits; expect tens of percent on top rather than another factor of several.

9.2 Route 2 — remove the Jordan–Wigner string with a fermionic swap network

The entire factor of N separating $P = \Theta(N^3)$ from $n_{\text{CNOT}} = \Theta(N^4)$ is the Z-string. Fermionic swap networks [8] remove it: by permuting the fermionic modes with FSWAP gates, any desired set of orbitals can be brought adjacent, where the excitation acts on four neighbouring qubits with no string at all and costs a flat 13 CNOTs (or 12 [4]). Kivlichan *et al.* show a full Trotter step of a general two-body Hamiltonian in depth N with $N^2/2$ two-qubit gates on a *linear* architecture.

The caveat, which should not be glossed over: their linear-depth result covers all *pairs* of orbitals, which suffices for a Hamiltonian written in a basis where the two-body term is a density–density interaction, not for an arbitrary set of $\Theta(N^3)$ four-index excitations. Routing $\Theta(N^3)$ distinct four-tuples through a linear array still costs $\Omega(N^3)$ FSWAPS in the worst case. The honest estimate is that swap networks convert the $\Theta(N^4)$ CNOT count into $\Theta(N^3)$ excitation gates plus a routing overhead that has to be determined by explicit network construction; the momentum-sector block structure (N independent sectors, each with $\Theta(N^2)$ generators) is the natural place to look for a good network, and is specific enough to this problem to be worth the effort. This is the least developed of the four routes and the one where a genuinely new result might be available.

9.3 Route 3 — compile the same amplitudes by double factorisation (recommended)

This subsection is self-contained: it states the idea, the objection to it, the answer, the recipe and the cost, without needing anything later in the note. Section A supplies the proofs and the momentum-space details for a reader who wants them.

9.3.1 What this route is, and what it is not

Routes 1 and 2 keep the circuit’s structure — one block per generator, P blocks in all — and make each block cheaper. Route 3 abandons that structure. It takes the optimised coefficients $A^{(2)}$ as *given* and asks for any circuit at all that implements the same unitary, however it is organised internally.

So this is a **compilation** problem, not a variational one. No wavefunction is re-optimised, no new ansatz is proposed, and the target is the manuscript’s own $\mathcal{U}_A$. The distinction matters, because the literature this route draws on — the unitary cluster Jastrow (UCJ) and local UCJ (LUCJ) ansätze [9, 10, 11] — normally presents the same mathematics the other way round, as a variational family to be optimised on hardware. Here the mathematics is used only as a change of circuit representation.

9.3.2 Two operators that have cheap circuits, and why

The $\Theta(N^4)$ of Section 6 came entirely from Jordan–Wigner strings: a physically four-body operator acquired a support of $\bar{w} \simeq (5N + 24)/9$ qubits. Any route to a cheaper circuit must therefore be built out of operators whose strings vanish. It is worth being explicit about when that happens, since it is the mechanism the whole route rests on.

Recall the transform, $\hat{c}_j = (\prod_{r < j} Z_r)^{\frac{1}{2}}(X_j + iY_j)$ and $\hat{c}_j^\dagger = (\prod_{r < j} Z_r)^{\frac{1}{2}}(X_j - iY_j)$. A string appears only when the Z -prefixes of the different ladder operators in a term fail to cancel. Two cases in which they cancel completely:

Number operators — at any separation. In $\hat{n}_p = \hat{c}_p^\dagger \hat{c}_p$ the two prefixes are *identical* and square to the identity, leaving

$$\hat{n}_p = \frac{1}{2}(1 - Z_p). \quad (31)$$

Hence $\hat{n}_p \hat{n}_q = \frac{1}{4}(1 - Z_p)(1 - Z_q)$ is a weight-two operator *however far apart p and q are*. A diagonal Coulomb operator $\hat{J} = \sum_{pq} J_{pq} \hat{n}_p \hat{n}_q$ therefore has no strings at all, for any J : it is a sum of commuting $Z_p Z_q$ and Z_p terms, so $e^{i\hat{J}}$ is a product of commuting CPhase gates — $M(M - 1)/2$ of them in depth $M - 1$ on an all-to-all device (edge-colouring the complete graph), or depth M with a swap network on a line.

Hopping between adjacent modes. In $\hat{c}_p^\dagger \hat{c}_{p+1}$ the prefixes differ by the single factor Z_p , and $(X_p - iY_p)Z_p = X_p - iY_p$ absorbs it, leaving

$$\hat{c}_p^\dagger \hat{c}_{p+1} - \text{h.c.} = \frac{i}{2}(X_p Y_{p+1} - Y_p X_{p+1}), \quad (32)$$

a weight-two $XX + YY$ operator — exactly the exchange interaction native to superconducting hardware, and the same primitive Cade *et al.* build their ansatz from (Section 8.1). Note the restriction: this works only for $|p - q| = 1$. A hopping term between distant modes still carries the full string.

Why that restriction is not a restriction for Gaussian unitaries. This is the non-obvious step. One might expect a general orbital rotation $e^{\hat{K}}$ to need hopping between distant modes, and hence strings. It does not: **any $M \times M$ unitary can be realised exactly as a network of $M(M - 1)/2$ nearest-neighbour Givens rotations, arranged in M layers** [8]. The long-range content of the rotation is carried by the *composition* of adjacent rotations, not by any single long-range gate. So every gate in the network is of the form Equation (32), string-free, and $e^{\hat{K}}$ costs $O(N^2)$ gates in depth $O(N)$.

Why the same trick fails for the two-body generators. A generator $\hat{c}_p^\dagger \hat{c}_q^\dagger \hat{c}_s \hat{c}_r$ has four distinct indices (Lemma 1), and relabelling the modes can bring at most one pair adjacent, not both — and a relabelling that helps one generator will hurt others. There is no fixed ordering that makes all P generators local, which is precisely the content of Section 5.4, and precisely why Route 2 has to *move* the modes with FSWAPS rather than merely order them once.

Summarising: both primitives cost $O(N^2)$ gates in $O(N)$ depth and neither pays the Jordan–Wigner penalty, against $16(w-1)$ CNOTs for a *single* two-body generator in Section 6. The question is what can be built from them.

9.3.3 One cheap factor already is an exact two-body exponential

Conjugation passes through an exponential, $X e^Y X^{-1} = e^{XYX^{-1}}$, so for unitary $e^{\hat{K}}$

$$\boxed{e^{\hat{K}} e^{i\hat{J}} e^{-\hat{K}} = \exp\left(i e^{\hat{K}} \hat{J} e^{-\hat{K}}\right) \equiv e^{\hat{B}}, \quad \hat{B} \equiv i e^{\hat{K}} \hat{J} e^{-\hat{K}}.} \quad (33)$$

This is an identity, not an approximation. The generator $\hat{B}$ is anti-Hermitian (because $\hat{J}$ is Hermitian and $e^{\hat{K}}$ unitary), and it is two-body and number-conserving (because conjugating by a Gaussian unitary preserves both properties). In other words $\hat{B}$ is *the same kind of object* as the manuscript’s $\sum_g A_g \hat{\tau}_g$.

That is the entire idea: the left-hand side is a cheap circuit — one Givens network, one layer of CPhase gates, one inverse Givens network, $O(N^2)$ gates in $O(N)$ depth — and the right-hand side says it exactly implements the exponential of a two-body anti-Hermitian generator. Everything remaining is bookkeeping: which generators are reachable this way, how many such factors are needed, and what is lost in stringing them together.

9.3.4 One factor is badly restricted — the objection

It is not true that one factor reaches an arbitrary two-body generator, and it is worth being precise about how badly it fails. Write $\tilde{\hat{n}}_p = e^{\hat{K}} \hat{n}_p e^{-\hat{K}}$ for the rotated number operators and diagonalise the real symmetric matrix $J = \sum_m \omega_m v^{(m)} (v^{(m)})^\top$. Then

$$e^{\hat{K}} \hat{J} e^{-\hat{K}} = \sum_{pq} J_{pq} \tilde{\hat{n}}_p \tilde{\hat{n}}_q = \sum_m \omega_m \hat{L}_m^2, \quad \hat{L}_m = \sum_p v_p^{(m)} \tilde{\hat{n}}_p. \quad (34)$$

So one factor generates a sum of squares of one-body Hermitian operators — and every $\hat{L}_m$ is a function of the *same* set $\{\tilde{\hat{n}}_p\}$, so they all **mutually commute**. A single factor therefore reaches only a *simultaneously diagonalisable* sum of squares: in plain terms, a density–density interaction viewed in a rotated single-particle basis. That is a vanishingly small subset of the two-body anti-Hermitian operators, which is precisely the objection one should raise against calling the two families equivalent.

9.3.5 The sum over factors is not restricted

The objection is answered by noting what the decomposition actually asks for: not one factor, but a sum over factors carrying *different, mutually non-commuting* rotations. That sum is unrestricted.

Proposition 4 (Exactness of the sum of squares). *Every two-body, number-conserving Hermitian operator $\hat{W}$ can be written*

$$\hat{W} = \sum_{\mu=1}^R \omega_{\mu} \hat{L}_{\mu}^2 + (\text{one-body}), \quad \omega_{\mu} \in \mathbb{R}, \quad (35)$$

with every $\hat{L}_{\mu}$ one-body Hermitian and $R \leq M(M+1)/2 = O(N^2)$.

Proof. Write $\hat{W} = \sum_{pqrs} W_{pq,rs} \hat{c}_p^{\dagger} \hat{c}_q^{\dagger} \hat{c}_r \hat{c}_s$ and regard W as a Hermitian matrix in the composite index (pq) . Diagonalising it gives $\hat{W} = \sum_{\mu} \omega_{\mu} \hat{L}_{\mu}^{\dagger} \hat{L}_{\mu}$ with real ω_{μ} and one-body, generally non-Hermitian, $\hat{L}_{\mu}$. Split $\hat{L}_{\mu} = \frac{1}{2}(X_{\mu} - iY_{\mu})$ into Hermitian parts; then $\hat{L}_{\mu}^{\dagger} \hat{L}_{\mu} = \frac{1}{4}(X_{\mu}^2 + Y_{\mu}^2) + \frac{1}{2}[\hat{L}_{\mu}^{\dagger}, \hat{L}_{\mu}]$, and the commutator of two one-body operators is one-body because $[\hat{c}_p^{\dagger} \hat{c}_q, \hat{c}_r^{\dagger} \hat{c}_s] = \delta_{qr} \hat{c}_p^{\dagger} \hat{c}_s - \delta_{sp} \hat{c}_r^{\dagger} \hat{c}_q$. The bound on R is the dimension of the symmetric part of the composite index. $\square$

So Equation (35) is an identity, not an ansatz, and the signs $\omega_{\mu} = \pm$ mean no positive-semidefiniteness is assumed: **there is no expressivity deficit at full rank**. Any deficit lives in truncating R , and nowhere else. Note also that the number of *circuit* factors is not R : by Equation (34) one factor absorbs an entire commuting group of the $\hat{L}_{\mu}$ at once.

9.3.6 The momentum-sector decomposition: how the construction actually looks here

Everything above would hold in any single-particle basis. It is now necessary to stop speaking generally, because this ansatz exists only in the momentum basis — the qubits *are* the momentum modes of Equation (3) — and the momentum structure is not a refinement of the construction but the form the construction takes. It determines how many factors there are, how they are grouped, what each one costs, and whether the lattice momentum survives.

The generator splits into N independent sectors. Momentum conservation forces $W_{pq,rs}$ to vanish unless $\mathbf{k}_p - \mathbf{k}_q = -(\mathbf{k}_r - \mathbf{k}_s) \equiv \mathbf{Q}$, the transferred momentum. Hence

$$\hat{W} = \sum_{\mathbf{Q}} \hat{W}_{\mathbf{Q}}, \quad (36)$$

and the composite-index matrix of Proposition 4 is **block diagonal in $\mathbf{Q}$** : N blocks of dimension $2N$, in place of one block of dimension $M^2 = 4N^2$. This is the same reduction from N^4 to N^3 independent momentum combinations that gives the manuscript its $\frac{3}{4}N^3$ parameter count, and it is the structure exploited in the Bloch-orbital double factorisation of Ref. [14]. Two consequences matter.

It caps the rank. $R \leq N \times 2N = 2N^2$, against $O(N^4)$ for a dense two-body operator on M modes. Practically, one never forms the large matrix: one stores and diagonalises N blocks of size $2N \times 2N$.

It is the grouping that preserves the symmetry. Each $\hat{W}_{\mathbf{Q}}$ transfers $+\mathbf{Q}$ in one ladder pair and $-\mathbf{Q}$ in the other, so its net momentum transfer is zero and $[\hat{W}_{\mathbf{Q}}, T] = 0$ for lattice translations T . Therefore $\prod_{\mathbf{Q}} e^{i\hat{W}_{\mathbf{Q}}}$ conserves $\mathbf{K}$ *exactly*, for any ordering of the sectors and regardless of the splitting error between them. Sector-by-sector is not one grouping among many: it is the one that keeps the quantum number the manuscript's ansatz is built around, at no cost.

Inside a sector: the squares come in $\pm\mathbf{Q}$ pairs. Within the $\mathbf{Q}$ block the natural Cholesky vector $\hat{L}_{n\mathbf{Q}}$ carries definite transfer $\mathbf{Q}$ and is therefore *not* Hermitian, so Proposition 4 must be applied through its Hermitian parts. Setting $\hat{A}_n = \hat{L}_{n\mathbf{Q}} + \hat{L}_{n\mathbf{Q}}^\dagger$ and $\hat{B}_n = i(\hat{L}_{n\mathbf{Q}} - \hat{L}_{n\mathbf{Q}}^\dagger)$,

$$\hat{W}_{\mathbf{Q}} = \frac{1}{2} \sum_n \omega_n (\hat{A}_n^2 + \hat{B}_n^2), \quad \hat{A}_n^2 + \hat{B}_n^2 = 2(\hat{L}_{n\mathbf{Q}}^\dagger \hat{L}_{n\mathbf{Q}} + \hat{L}_{n\mathbf{Q}} \hat{L}_{n\mathbf{Q}}^\dagger). \quad (37)$$

Individually $\hat{A}_n^2$ has components at transfer 0 and $\pm 2\mathbf{Q}$ and does *not* conserve $\mathbf{K}$; the $\pm 2\mathbf{Q}$ pieces cancel only in the pair. Since $\hat{A}_n$ and $\hat{B}_n$ have different diagonalising rotations they become two separate circuit factors, so the within-sector splitting reintroduces a momentum-violating remainder — at $O(\beta^2)$ with a plain ordering, or $O(\beta^3)$ if each $\pm\mathbf{Q}$ pair is split symmetrically, $e^{i\theta\hat{A}^2/2} e^{i\theta\hat{B}^2} e^{i\theta\hat{A}^2/2}$. The leakage is therefore bounded by the splitting error, not independent of it, and it is directly measurable as $1 - \|P_{\mathbf{K}} \mathcal{U}_{\text{compiled}} |\Phi_{\mathbf{K}}\rangle\|^2$. Throughout, $N_\uparrow$, $N_\downarrow$ and S^z are *exactly* conserved, since every factor is number-conserving by construction.

The rotations inherit the cycle structure of $\mathbf{Q}$. The one-body matrix of $\hat{A}_n(\mathbf{Q})$ is supported only on the entries $(\mathbf{k}, \mathbf{k} \ominus \mathbf{Q})$ and their transposes. Repeatedly subtracting $\mathbf{Q}$ partitions the Brillouin zone into cycles of length $m_{\mathbf{Q}} = \text{ord}(\mathbf{Q})$, so in a basis ordered along those cycles $\hat{A}_n(\mathbf{Q})$ is block-diagonal with $N/m_{\mathbf{Q}}$ cyclic-tridiagonal blocks of size $m_{\mathbf{Q}}$ (per spin). Its diagonalising rotation inherits that block structure, costing $O(Nm_{\mathbf{Q}})$ Givens rotations in depth $O(m_{\mathbf{Q}})$ — against $O(N^2)$ in depth $O(N)$ for a generic orbital rotation. Three regimes:

- $\mathbf{Q} = 0$: $m = 1$, the operator is already diagonal, and the **entire forward-scattering sector collapses to a single factor with no rotation at all** — one CPhase layer.
- $\text{ord}(\mathbf{Q}) = 2$: disjoint 2×2 blocks, $O(N)$ gates in depth 1.
- generic $\mathbf{Q}$, $\text{ord}(\mathbf{Q}) \sim N$: no saving over a dense rotation.

Most $\mathbf{Q}$ fall in the third regime, so the rotations are *not* asymptotically cheaper than generic ones; what the cycle structure buys is a constant. Summed over all factors the rotations cost between 11% and 54% of the CPhase layers on the clusters of Table 2, so **the diagonal Coulomb layer sets the cost** and the estimates below are governed by it.

The Hubbard vertex is exactly one factor. Let F be the Gaussian unitary implementing the fermionic Fourier transform between the momentum modes carried by the qubits and real-space modes. Then

$$U \sum_j \hat{n}_{j\uparrow} \hat{n}_{j\downarrow} = F \hat{J} F^\dagger, \quad \hat{J} = U \sum_j \hat{n}_{j\uparrow} \hat{n}_{j\downarrow} \text{ diagonal in the qubit basis}, \quad (38)$$

so $e^{i\theta\hat{V}}$ is exactly one factor of the form Equation (33), at $k = 1$ and with zero error. The method therefore contains, at the lowest possible rank, the very interaction it is being asked to dress — which is the strongest available reason to expect the rank truncation of option (ii) below to behave well here. It also means that, since the kinetic term is one-body and free, a Trotter step of the full Hubbard Hamiltonian is one orbital rotation plus one factor: **the Hamiltonian Variational ansatz is the special case of this family in which every $\hat{K}_\mu$ is fixed to F and every $\hat{J}_\mu$ to the on-site pattern**, so the two ansätze compared in Section 8 are members of one family rather than competitors.

9.3.7 The recipe, sector by sector

The ansatz being compiled is the product form of Equation (1), at $L = 1$,

$$\mathcal{U}_A = \prod_{g=1}^P e^{A_g \hat{\tau}_g}, \quad (39)$$

with a fixed ordering of the P factors — *not* the single exponential $e^{\sum_g A_g \hat{\tau}_g}$. The distinction matters at the outset, because double factorisation consumes a *sum*. Step 0 therefore has to say which generator is being fed in, and it is the step most easily got wrong.

Step 0 — choose the input coefficients. Three options, in increasing order of rigour and cost.

- (a) **Use the single-exponential coefficients, which already exist.** The manuscript optimises both forms and compares them in Fig. 4; the single-exponential runs (the unregularised “noreg” coefficients) define $\hat{A} = \sum_g A_g \hat{\tau}_g$ unambiguously, which is exactly what double factorisation consumes. **This is the right input and requires no new optimisation at all.**
- (b) **Re-sum the product-form coefficients.** Taking the A_g from a Trotter-form run and forming $\hat{A} = \sum_g A_g \hat{\tau}_g$ is tempting but not free: those coefficients were optimised against a specific ordering of P factors, and the sum differs from that product at $O(\beta^2)$. The resulting circuit validly compiles a *different* unitary from the one whose fidelity was measured, so the fidelity must be re-checked — which Step 6 does anyway, so this option is viable, just not free.
- (c) **Re-optimize with the compiled circuit in the loop.** Most rigorous and strictly better than compiling fixed coefficients, but it is a new optimisation and turns the exercise back into a variational one.

Take (a). Write the resulting generator as $\hat{A} = i\hat{W}$ with $\hat{W}$ Hermitian.

Step 1 — sort the generators into momentum sectors. Each generator is $\hat{\tau}_{IJ} = \hat{c}_p^\dagger \hat{c}_q^\dagger \hat{c}_s \hat{c}_r - \text{h.c.}$ with $I = (p, q)$, $J = (r, s)$. Normal-order it into $\sum W_{pq,rs} \hat{c}_p^\dagger \hat{c}_q \hat{c}_r^\dagger \hat{c}_s$ using $\hat{c}_p^\dagger \hat{c}_q^\dagger \hat{c}_s \hat{c}_r = -\hat{c}_p^\dagger \hat{c}_s \hat{c}_q^\dagger \hat{c}_r + \delta_{qs} \hat{c}_p^\dagger \hat{c}_r$, collecting the δ term as a one-body remainder to be applied as a single orbital rotation at the end. Bin each contribution by its transferred momentum $\mathbf{Q} = \mathbf{k}_p - \mathbf{k}_q$, giving N Hermitian matrices $W_{\mathbf{Q}}$ of size $2N \times 2N$ — never a dense $4N^2 \times 4N^2$ array. **Exact.**

Step 2 — first factorisation, per sector. Diagonalise each $W_{\mathbf{Q}}$, giving $\hat{W}_{\mathbf{Q}} = \sum_n \omega_n \hat{L}_n^\dagger \hat{L}_n$, and form the Hermitian pairs $\hat{A}_n, \hat{B}_n$ of Equation (37). For 4×3 this is twelve 24×24 eigendecompositions — milliseconds. **Exact.**

Step 3 — second factorisation. Diagonalise each Hermitian $\hat{A}_n, \hat{B}_n$ by its own orbital rotation, exploiting the $\mathbf{Q}$ -cycle block structure so that only blocks of size $m_{\mathbf{Q}}$ are diagonalised. Output per factor: $(\kappa_\mu, \lambda^\mu, \omega_\mu)$ with $J_{pq}^\mu = \omega_\mu \lambda_p^\mu \lambda_q^\mu$. **Exact.** These two stages are what “double factorisation” names.

Step 4 — split the exponential. Order the factors **sector by sector**, $\prod_{\mathbf{Q}} \prod_n$, and within each $\pm \mathbf{Q}$ pair use the symmetric ordering $e^{i\theta \hat{A}_n^2/2} e^{i\theta \hat{B}_n^2} e^{i\theta \hat{A}_n^2/2}$. **This is the only approximation in the route**, and this ordering is what keeps the momentum leakage at $O(\beta^3)$ rather than $O(\beta^2)$.

Step 5 — emit the circuit. For each factor: the Givens network for $e^{\hat{K}_\mu}$ (Equation (32) gates), the CPhase layer for $e^{i\hat{J}_\mu}$ (Equation (31) gates), the inverse Givens network. Skip the rotations entirely for the $\mathbf{Q} = 0$ sector. Append the one-body remainder from Step 1 as a final orbital rotation. Nothing here carries a Jordan–Wigner string.

Step 6 — validate classically, with the existing loss. Apply the emitted circuit to $|\Phi_{\mathbf{K}}\rangle$ as a state-vector operation and evaluate $1 - |\langle \Psi_{\mathbf{K}}(U) | \mathcal{U}_{\text{compiled}} | \Phi_{\mathbf{K}} \rangle|^2$ — the manuscript’s own loss on its own reference and target states, so the answer is directly comparable to the published fidelities and no new machinery is needed. Evaluate the momentum leakage $1 - \|P_{\mathbf{K}} \mathcal{U}_{\text{compiled}} | \Phi_{\mathbf{K}} \rangle\|^2$ at the same time; it is one projection.

Step 7 — optional: compress. Replace the exact eigendecomposition of Step 2 by a rank- k fit under the unitary constraint [12], and repeat Steps 3–6 as a function of k . This is the measurement recommended below.

A knob: how coarsely to group. Steps 0–5 sum all P generators before factorising; Equation (39) keeps all P separate. Between them, one may sum within groups of sectors and factorise each group. The sector-resolved choice above is the natural stopping point — it is where the momentum quantum number is exactly preserved and where the first factorisation loses nothing — but coarser groupings reduce the splitting error further and are free to scan.

9.3.8 Why the one approximation is benign

Step 4 is an ordinary first-order Trotter splitting, with error $\frac{1}{2} \sum_{\mu < \nu} \|[\hat{B}_\mu, \hat{B}_\nu]\| = O(R^2 \beta^2)$ where $\beta = \max_\mu \|\hat{B}_\mu\|$ — second order in the amplitudes, and reducible to third order by a symmetric ordering at twice the gate count.

It is important that this is *not a new class of error*. Equation (39) is itself a first-order Trotter splitting of $e^{\hat{A}}$ — into $P = \frac{3}{4}N^3$ factors. Step 4 replaces it with a splitting into $R \leq 2N^2$ factors: at 4×3 , 288 factors in place of 1092. A coarser grouping into fewer, larger blocks does not generally increase Trotter error, and the manuscript supplies direct evidence that this order of error is not limiting, since in Fig. 4 the product form is the *more* accurate of the two at small $\dim \mathcal{H}$. The route therefore trades one Trotterisation for a coarser one; it does not trade an exactness for an approximation.

9.3.9 Cost: a guaranteed figure and a conjectural one

These must be kept apart.

(i) **Exact rank: guaranteed.** With $R \leq 2N^2$ factors at $\Theta(N^2)$ gates and $\Theta(N)$ depth each,

$$n_{2q}^{\text{exact DF}} = \Theta(N^4), \quad D^{\text{exact DF}} = \Theta(N^3). \quad (40)$$

The gate count is not improved asymptotically, only by a constant of about $1.4\times$; the depth improves by a factor of N . No assumption enters.

(ii) **Compressed rank: conjectural.** Fitting rather than truncating to rank k gives $O(kN^2)$ gates in $O(kN)$ depth — the $O(kN^2)$ scaling reported for the k -fold UCJ family. Parameter counting permits $k = \Theta(N)$, since $kN^2 \sim N^3 \sim P$, which would give $\Theta(N^3)$ gates in $\Theta(N^2)$ depth. Whether the manuscript’s particular $A^{(2)}$ admits such a compression is exactly the open question. Rubin, Lee and Babbush give the algorithm that answers it: rather than truncating a spectrum, it fits $\{\hat{K}_\mu, \hat{J}_\mu\}$ to $\hat{W}$ directly under the unitary constraint, and reports “many times fewer” squared one-body operators than the analytic decomposition [12]. The local variant restricts J_μ to a hardware-native connectivity pattern, removing SWAP overhead [11].

cluster	gates / depth				
	staircase	Yordanov	(i) exact DF, $R = 2N^2$	(ii) compressed DF, $k = N$	
4×3	$1.50 \times 10^5 / 1.24 \times 10^5$	$2.64 \times 10^4 / 1.69 \times 10^4$	$1.07 \times 10^5 / 8.9 \times 10^3$	$4.5 \times 10^3 / 3.7 \times 10^2$	
4×4	$4.67 \times 10^5 / 3.85 \times 10^5$	$7.73 \times 10^4 / 5.10 \times 10^4$	$3.19 \times 10^5 / 2.0 \times 10^4$	$1.0 \times 10^4 / 6.2 \times 10^2$	
6×6	$1.16 \times 10^7 / 9.42 \times 10^6$	$1.68 \times 10^6 / 1.20 \times 10^6$	$7.37 \times 10^6 / 2.0 \times 10^5$	$1.0 \times 10^5 / 2.8 \times 10^3$	

Read columns (i) and (ii) as a floor and a target. Even the floor is a 14× depth reduction at 4×3 and 46× at 6×6 relative to the staircase, at a *lower* gate count. Against the Yordanov circuits of Route 1 the floor is 1.9× shallower but uses 4× more gates, so the two routes win on different axes and the choice depends on whether the device is depth- or gate-limited. The target, if achievable, beats both by more than an order of magnitude on each axis.

9.3.10 The experiment this implies

Take the optimised $A^{(2)}$ from an existing 4×3 or 4×4 run, compress it with the Rubin *et al.* algorithm, and plot infidelity against k .

Three properties make this unusually cheap and unusually informative. It requires no new wavefunction optimisation, because the coefficients are already in hand. Its $k \rightarrow R$ endpoint is pinned at the manuscript’s own fidelity by Proposition 4, so the curve is guaranteed to terminate correctly and the only thing being measured is how fast it gets there. And the curve itself *is* the accuracy-versus-depth trade-off, which is the quantity the paper currently lacks.

If it flattens by $k \sim N$, column (ii) is real and the ansatz is a near-term circuit proposal. If it does not, column (i) still buys a factor of N in depth, and the honest conclusion is that this is a classical-structure result — which is in any case how the manuscript currently frames its hardware outlook.

9.4 Route 4 — shrink the generator set

Orthogonal to the compilation, and multiplicative with it:

- **Impose $SU(2)$.** Appendix C of the manuscript derives but does not impose the singlet constraint. Since the count is quoted there as $\sim \frac{1}{4}N^3$ against $\frac{3}{4}N^3$, this is a direct factor ≈ 3 off every number in Table 2, at the price of needing a total-spin-eigenstate reference.
- **Amplitude pruning.** The workflow already prunes the smallest coefficients. The gate count is exactly linear in the number of surviving generators, so a measured pruning fraction translates one-to-one into a circuit saving — worth reporting alongside the fidelity, as it is the cheapest quantitative bridge from the classical study to a hardware claim.
- **ADAPT-style selection.** Growing the layer operator-by-operator by gradient magnitude typically reaches comparable accuracy with far fewer generators than the full set; this is the standard alternative to pruning a dense $A^{(2)}$ and is the setting the Yordanov circuits were designed for.

- **Momentum-space UCC precedent.** The reduction of the excitation set by momentum conservation, and the associated circuit savings, are treated for periodic systems in Ref. [15]; worth citing as prior art for the $N^4 \rightarrow N^3$ parameter reduction claimed in the manuscript.

10 Summary and recommendations

1. **Correct the parameter count** to $P = \frac{1}{4}(3N^3 - 6N^2 + (g + 2)N)$, $g = 2^{\#\{d:L_d \text{ even}\}}$ (Proposition 1). The quoted $\frac{1}{4}(3N^3 + 2N^2)$ counts $2N^2 - N$ identically vanishing diagonal generators and assumes $g = 2$. The corrected value reproduces the 114 parameters used in the 3×2 runs; at 4×3 it is 1092, not 1368. The $\frac{3}{4}N^3$ asymptotic scaling, which is what the paper’s argument rests on, is unaffected.
2. **The naïve circuit is $\Theta(N^4)$ cnots and $\Theta(N^4)$ depth**, not $\Theta(N^3)$: $n_{\text{CNOT}} \simeq \frac{20}{3}N^4$, concretely 1.5×10^5 at 4×3 and 4.7×10^5 at 4×4 . The extra factor of N over the parameter count is purely the Jordan–Wigner string.
3. **Parallelism cannot fix this.** Each generator occupies a fraction $\bar{w}/2N \rightarrow 5/18$ of the register, capping concurrency at 3.6 regardless of N ; a greedy schedule realises only $1.23\times$. The premise of the question is correct and now has a number attached to it.
4. **Against Cade *et al.*** (Section 8), whose EHV circuit reaches 0.99 fidelity in $\Theta(N^{3/2})$ two-qubit depth — 108 layers at 4×3 — one UCC layer is $1100\times$ deeper, $160\times$ after Yordanov recompilation, $8\times$ even in doubly-factorised form. The two ansätze have opposite locality structure, and the manuscript’s parameter-count advantage does not survive translation into depth. The experiment that would settle it — infidelity versus compiled two-qubit depth for both ansätze on one cluster — is now within reach in this codebase.
5. **Free depth win:** balanced parity trees take the depth from $\Theta(N^4)$ to $\Theta(N^3 \log N)$ with no change in gate count and no change to the ansatz.
6. **Best compilation of the same ansatz:** Yordanov excitation circuits, $8\times$ fewer CNOTs and $\approx 7.5\times$ shallower.
7. **Best overall route:** compile the *same* amplitudes by double factorisation (Section 9.3). A single UCJ factor reaches only rotated density–density operators, so the two families are not equivalent factor-by-factor; but the sum over factors with non-commuting rotations is exact at rank $R \leq M(M + 1)/2$ (Proposition 4), so there is no expressivity deficit, and the only approximation is a Trotter splitting into $R \leq 2N^2$ factors where the manuscript already accepts one into $P = \frac{3}{4}N^3$. Unconditionally this gives $\Theta(N^4)$ gates and $\Theta(N^3)$ depth; compressed to $k = N$ it would give $\Theta(N^3)$ and $\Theta(N^2)$. The immediate next step is to compress an already-optimised $A^{(2)}$ with the Rubin *et al.* algorithm and measure infidelity against k — a curve pinned at its $k \rightarrow R$ end by Proposition 4.

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A Double factorisation in the momentum basis: derivation and error orders

Section 9.3 stated the compilation and its cost. This appendix supplies the derivations: the factor identity, what a single factor generates, the sector-resolved factorisation of the generator, the cycle structure of the rotations, and the order of every error. It is written in the momentum basis throughout — not as a specialisation of a general construction, since the ansatz being compiled exists in no other basis — and it can be read in order from Section A.1.

A.1 Setup: the momentum basis

The qubits are the $M = 2N$ momentum modes $(\mathbf{k}, \sigma)$ of Equation (3). The object to be compiled is $e^{\hat{A}}$ with $\hat{A} = i\hat{W}$ and $\hat{W} = \sum_{pqrs} W_{pq,rs} \hat{c}_p^\dagger \hat{c}_q \hat{c}_r^\dagger \hat{c}_s$ two-body, Hermitian and number-conserving — the manuscript's $\sum_g A_g \hat{\tau}_g$ written in that form. Two operator families appear throughout:

$$\hat{K} = \sum_{pq} \kappa_{pq} \hat{c}_p^\dagger \hat{c}_q \quad (\kappa^\dagger = -\kappa), \quad \hat{J} = \sum_{pq} J_{pq} \hat{n}_p \hat{n}_q \quad (J \in \mathbb{R}^{M \times M} \text{ symmetric}), \quad (41)$$

with $\hat{n}_p = \hat{c}_p^\dagger \hat{c}_p$. By Equation (31) and Equation (32), $e^{i\hat{J}}$ is a layer of commuting CPhase gates and $e^{\hat{K}}$ a network of nearest-neighbour Givens rotations, both free of Jordan–Wigner strings.

A.2 The factor identity, and what one factor generates

The identity. For invertible X and any Y , $Xe^Y X^{-1} = e^{XYX^{-1}}$, since conjugation is an algebra homomorphism and may be applied term by term to the exponential series. With $X = e^{\hat{K}}$ unitary this is Equation (33). The generator $\hat{B} = i e^{\hat{K}} \hat{J} e^{-\hat{K}}$ is anti-Hermitian because $\hat{J}$ is Hermitian and $e^{\hat{K}}$ unitary, and two-body and number-conserving because conjugation by a Gaussian unitary maps $\hat{c}_p^\dagger \mapsto \sum_a U_{ap} \hat{c}_a^\dagger$ and so preserves both particle number and the rank of a normal-ordered operator.

What it generates. With $\tilde{\hat{c}}_p^\dagger = e^{\hat{K}} \hat{c}_p^\dagger e^{-\hat{K}} = \sum_a U_{ap} \hat{c}_a^\dagger$, $U = e^\kappa$, and $\tilde{\hat{n}}_p = \tilde{\hat{c}}_p^\dagger \tilde{\hat{c}}_p$, one has $e^{\hat{K}} \hat{J} e^{-\hat{K}} = \sum_{pq} J_{pq} \tilde{\hat{n}}_p \tilde{\hat{n}}_q$; diagonalising $J = \sum_m \omega_m v^{(m)} (v^{(m)})^\top$ gives Equation (34). Every $\hat{L}_m$ is a function of the same commuting set $\{\tilde{\hat{n}}_p\}$, hence they commute — the restriction discussed in Section 9.3. Explicitly the generated tensor is

$$W_{ab,cd} = \sum_m \omega_m \ell_{ab}^{(m)} \ell_{cd}^{(m)}, \quad \ell_{ab}^{(m)} = \sum_p v_p^{(m)} U_{ap} U_{bp}^*, \quad (42)$$

an outer product in the composite index whose factors are themselves outer products in the orbital index — the “twice-factorised” form the method is named for.

A.3 The first factorisation, sector by sector

Crystal momentum conservation forces $W_{pq,rs} = 0$ unless $\mathbf{k}_p - \mathbf{k}_q = -(\mathbf{k}_r - \mathbf{k}_s) \equiv \mathbf{Q}$, so W is block diagonal in $\mathbf{Q}$ (Equation (36)) with N blocks of dimension $2N$. Each block is diagonalised separately:

$$W_{pq,rs}^{\mathbf{Q}} = \sum_n \omega_n \ell_{pq}^{n\mathbf{Q}} (\ell_{rs}^{n\mathbf{Q}})^* \quad \implies \quad \hat{W}^{\mathbf{Q}} = \sum_n \omega_n \hat{L}_{n\mathbf{Q}}^\dagger \hat{L}_{n\mathbf{Q}}, \quad \hat{L}_{n\mathbf{Q}} = \sum_{pq} \ell_{pq}^{n\mathbf{Q}} \hat{c}_p^\dagger \hat{c}_q, \quad (43)$$

which is a Cholesky decomposition when the block is positive semidefinite and an eigendecomposition in general, the ω_n then carrying signs. Summing the block ranks gives $R \leq N \times 2N = 2N^2$, against $O(N^4)$ for a dense two-body operator on M modes. Since $[\hat{W}^{\mathbf{Q}}, T] = 0$ for lattice translations T , every sector generator conserves $\mathbf{K}$ exactly — the point made in Section 9.3.6.

Hermitisation, and why the squares pair. Proposition 4 requires Hermitian $\hat{L}$, but $\hat{L}_{n\mathbf{Q}}$ carries definite transfer $\mathbf{Q}$ and is Hermitian only at $\mathbf{Q} = 0$. Passing to Hermitian parts $\hat{A}_n, \hat{B}_n$ gives Equation (37). Individually $\hat{A}_n^2$ contains components at transfer 0 and $\pm 2\mathbf{Q}$; the latter cancel only against those of $\hat{B}_n^2$. Two consequences: the pair must be kept together to conserve $\mathbf{K}$,

and since $\hat{A}_n$ and $\hat{B}_n$ do not commute in general they cannot be merged into one circuit factor, so within-sector splitting is unavoidable. A symmetric ordering of each pair holds the resulting leakage to $O(\beta^3)$.

A.4 The second factorisation and the cycle structure of the rotations

Each Hermitian $\hat{A}_n(\mathbf{Q})$ is one-body, hence diagonalised by an orbital rotation:

$$\hat{A}_n = e^{\hat{K}_\mu} \left(\sum_p \lambda_p^\mu \hat{n}_p \right) e^{-\hat{K}_\mu} \implies \omega_\mu \hat{A}_n^2 = e^{\hat{K}_\mu} \hat{J}_\mu e^{-\hat{K}_\mu}, \quad J_{pq}^\mu = \omega_\mu \lambda_p^\mu \lambda_q^\mu. \quad (44)$$

The momentum structure constrains that rotation sharply.

The matrix of $\hat{A}_n(\mathbf{Q})$ is supported only on the entries $(\mathbf{k}, \mathbf{k} \ominus \mathbf{Q})$ and their transposes. Iterating $\mathbf{k} \mapsto \mathbf{k} \ominus \mathbf{Q}$ partitions the Brillouin zone into cycles of length $m_{\mathbf{Q}} = \text{ord}(\mathbf{Q})$, the order of $\mathbf{Q}$ in $\mathbb{Z}_{L_1} \times \mathbb{Z}_{L_2}$, namely $m_{\mathbf{Q}} = \text{lcmd}(L_d / \text{gcd}(Q_d, L_d))$. Ordered along those cycles, $\hat{A}_n(\mathbf{Q})$ is block diagonal with $N/m_{\mathbf{Q}}$ cyclic-tridiagonal blocks of size $m_{\mathbf{Q}}$ per spin. Its eigenvectors are confined to a single cycle but are generically dense *within* it, so the diagonalising rotation is block diagonal with blocks of size $m_{\mathbf{Q}}$, costing $O(Nm_{\mathbf{Q}})$ Givens rotations in depth $O(m_{\mathbf{Q}})$ once the disjoint blocks are run in parallel. This is cheaper than a generic $O(N^2)$ -gate, $O(N)$ -depth rotation by a factor $N/m_{\mathbf{Q}}$, but only for small $m_{\mathbf{Q}}$: at $\mathbf{Q} = 0$ the operator is already diagonal and no rotation is needed at all, at $\text{ord}(\mathbf{Q}) = 2$ the blocks are 2×2 , and for generic $\mathbf{Q}$ with $m_{\mathbf{Q}} \sim N$ there is no saving. Since most $\mathbf{Q}$ have large order the gain is a constant, not a scaling: summed over all factors the rotations cost 11%–54% of the CPhase layers on the clusters of Table 2, so the diagonal Coulomb layers set the cost.

A.5 The splitting error

The steps above are exact; the approximation is in turning the sum into a product. By Baker–Campbell–Hausdorff,

$$\prod_{\mu=1}^k e^{\hat{B}_\mu} = \exp \left(\sum_{\mu} \hat{B}_\mu + \frac{1}{2} \sum_{\mu < \nu} [\hat{B}_\mu, \hat{B}_\nu] + O(\beta^3) \right), \quad \beta \equiv \max_{\mu} \|\hat{B}_\mu\|, \quad (45)$$

so

$$\left\| \prod_{\mu} e^{\hat{B}_\mu} - e^{\sum_{\mu} \hat{B}_\mu} \right\| \leq \frac{1}{2} \sum_{\mu < \nu} \|[\hat{B}_\mu, \hat{B}_\nu]\| + O(k^3 \beta^3) = O(k^2 \beta^2), \quad (46)$$

ordinary first-order Trotter error, second order in the amplitudes and reducible to $O(k^3 \beta^3)$ by symmetric ordering at twice the gate count. Two structural remarks.

- **It is variationally absorbable.** If $\{\hat{K}_\mu, \hat{J}_\mu\}$ are re-optimised rather than read off, Equation (46) bounds the distance to a *particular* target, not the distance from the variational manifold to the eigenstate; re-optimisation can only improve on the compiled circuit.
- **Symmetry is a question of grouping, not of Trotter order.** If every factor commutes with a symmetry generator, so does the product, exactly and at every order in Equation (45). This is why the sector grouping of Section 9.3.6 is worth adopting: it makes $\mathbf{K}$ exact between sectors, leaving only the within-sector $\pm \mathbf{Q}$ pairing to contribute leakage, at $O(\beta^3)$ under symmetric ordering.

A.6 The Hubbard vertex, and HVA as a special case

Let F be the Gaussian unitary implementing the fermionic Fourier transform between the momentum modes carried by the qubits and real-space modes. Then Equation (38) holds: $e^{i\theta\hat{V}}$ is exactly one factor, with $e^{\hat{K}} = F$, at $k = 1$ and zero error. Three consequences, of which the first two are stated in Section 9.3.6:

1. The method contains the interaction it is asked to dress at the lowest possible rank, so the rapid-decay assumption behind rank truncation has no generic reason to fail here.
2. Since the kinetic term is one-body and free, a Trotter step of the full Hubbard Hamiltonian is one orbital rotation plus one factor, making the Hamiltonian Variational ansatz the special case in which every $\hat{K}_\mu$ is fixed to F and every $\hat{J}_\mu$ to the on-site pattern. The ansätze compared in Section 8 are members of one family.
3. Because $\hat{J}$ in Equation (38) is real-space diagonal, it is a specific combination across *all* $\mathbf{Q}$ at once. Momentum-conserving structure in this basis is natural when organised by real-space locality and awkward when organised $\mathbf{Q}$ by $\mathbf{Q}$ — the pairing problem of Section A.3 seen from the other side. It also indicates the fallback if the measured leakage is large: restrict to $e^{\hat{K}_\mu} = F e^{\hat{k}_\mu}$ with $\hat{k}_\mu$ a translation-invariant real-space one-body operator and $J_\mu(j, j') = J_\mu(j - j')$, a family in which every factor conserves $\mathbf{K}$ exactly at every order, at the price of $O(N)$ rather than $O(N^2)$ parameters per factor.

A.7 Error budget

source	where	order	controlled by
factor identity Equation (33)	§A.3	exact	—
first + second factorisation Equation (43)	§A.3	exact at full rank $R \leq 2N^2$	—
rank truncation to $k < R$	§A.3	$\sum_{\mu > k} \omega_\mu \ \hat{L}_\mu\ ^2$	eigenvalue decay of $W_{\mathbf{Q}}$; compressed fitting
product vs. sum Equation (45)	Equation (46)	$O(k^2\beta^2)$: second order in the amplitudes	Trotter order; re-optimisation
$\mathbf{K}$ leakage, between sectors	Section 9.3.6	zero	sector grouping
$\mathbf{K}$ leakage, within a sector	Equation (37)	$O(\beta^3)$ with symmetric $\pm\mathbf{Q}$ ordering	pair ordering; translation-covariant fallback
$N_\uparrow, N_\downarrow, S^z$	—	exact	every factor is number-conserving
locality truncation (LUCJ)	Section 9.3	empirical	connectivity; reported exact at $L > 1, 2, 3, 4$ for all-to-all, square, hex, heavy-hex [11]

Of the eight entries, three are exact by construction, one is zero because of the sector grouping, and only three are genuine approximations — all of second or third order, and all attenuated in practice: the splitting error because the parameters may be re-optimised rather than imported, the leakage because symmetric pair ordering raises it an order, and the truncation error because compressed double factorisation fits $\{\hat{K}_\mu, \hat{J}_\mu\}$ to $\hat{W}$ directly under the unitary constraint instead of truncating a spectrum, which Ref. [12] reports needs “many times fewer” squared one-body operators than the analytic decomposition. That is the measurement recommended at the end of Section 9.3.

B Reproduction

All enumerated numbers in this note come from a direct construction of the generator set and an explicit Jordan–Wigner transform (no external libraries), followed by greedy list-scheduling on the conflict graph. The scripts are in `gate_count_scripts/`: `enumerate.py` (generator set and parameter counts), `jw.py` (Pauli algebra and JW transform, validated against Eq. (10)), `gates2.py`–`gates4.py` (weights, gate counts, schedules, circuit-model comparison), `order.py` (qubit-ordering study), `cade_cmp.py` (the comparison of Section 8; it reproduces the $24 + 25 \times 12 + 1 = 325$ worked example of Ref. [13] as a check on the convention). The parameter formula (8) was checked against brute-force enumeration for all clusters $L_x \times L_y$ with $N \leq 48$, and the per-generator cost (12) against the independently published $16(n^{(df)} - 1)$ of Ref. [2].